

Virginia Post-Incarceration Action Hub

Integrating correctional data with regional labor market analytics is critical for public policy, specifically to reduce recidivism and improve economic mobility for justice-involved populations. The Virginia Post-Incarceration Economic Opportunity Map & Action Hub acts as a computational framework to address barriers facing the ~10,800 individuals released from state incarceration in the Commonwealth annually. Although Virginia achieved a nationally recognized three-year re-incarceration rate of 17.6% for the Fiscal Year 2020 cohort—the lowest among 34 reporting states—the economic burden remains high, with recidivism costing the state approximately \$496.8 million in 2021.¹ This report analyzes the Hub's technical architecture, the statistical methodologies used for data interpretation, and the policy logic driving its recommendations.

Technical Architecture and Website Construction

The Virginia Post-Incarceration Economic Opportunity Map & Action Hub synthesizes heterogeneous datasets into a unified, interactive geospatial environment. Its architecture uses a multi-layered map to overlay correctional release data directly with economic development zones, workforce areas, and educational infrastructure. This way, prospective employees or employers can quickly find information local to their region and see how it relates to nearby counties/regions.

Geospatial Layering and Administrative Boundaries

The Hub's architecture is anchored by the nine GO Virginia regions certified by the Virginia Growth and Opportunity Board.⁴ These regions function as the primary units for analyzing collaboration across business, education, and government. By mapping these boundaries against the fourteen Local Workforce Development Areas (LWDAs) and twenty-one Planning District Commissions (PDCs), the system reveals granular economic realities rather than a generalized overview.⁵

Technically, the build process integrated shapefiles and FIPS-coded data from the Virginia Economic Development Partnership (VEDP) and the Department of Housing and Community Development (DHCD).⁴ This integration enables localized economic profiling, spanning from the Appalachian-focused Region One to the high-tech corridors of Northern Virginia's Region Seven.⁴

Data Engineering and Integration

The Hub's back-end operates as a relational database environment, synchronizing streams from the Virginia Department of Corrections (VADOC), Virginia Works (Department of Workforce

Development and Advancement), and the Bureau of Labor Statistics (BLS).⁸ During engineering, temporal misalignment emerged as a critical hurdle. VADOC recidivism data demands a four-year maturation period for the VirginiaCORIS system to fully capture court information.¹ Conversely, labor market datasets like Current Employment Statistics (CES) and the Job Openings and Labor Turnover Survey (JOLTS) update monthly.⁹

To bridge this gap, the architecture runs a data-normalization pipeline. This standardizes 'Gold Standard' historical recidivism cohorts against 'Predictive' real-time labor demand indicators. The result is a dynamic 'Reentry Success Potential' score, derived by weighing regional reentering population density against 'Second-Chance Friendly' vacancies in sectors such as manufacturing (NAICS 31-33), construction (NAICS 23), and transportation/warehousing (NAICS 48-49).⁸

Front-End Interface and User Experience

The front-end 'Action Hub' operates on a 'No Wrong Door' philosophy, granting reentrants, employers, and service providers zero-latency access to critical toolkits.¹⁰ Core features include a 'College Locator' and the interactive 'Network2Work' directory, which leverages geolocation to instantly map users to the nearest of Virginia's 23 community colleges or local 'Connectors'.¹⁵ Under the hood, the stack handles bandwidth-intensive utilities like video 'Job Tickets' and dynamic needs surveys, pinpointing specific friction points such as childcare or transportation gaps.¹⁷

Methodology & Data Engineering

Estimating the Returnee Flow

The Virginia Department of Corrections (VADOC) does not publish predictive datasets for post-release residency. We solved this by using a 'Stock-to-Flow' proxy method.⁶ We identified the home origin of current inmates (the 'Stock') using Prison Policy Initiative (PPI) 2020 Census data. We then converted this static figure into an annual 'Flow' of returnees. We applied a density-based formula using VADOC's baseline release volume of 12,680 individuals.¹

Formula:

$$\text{Est. Annual Returnees}_c = \left(\frac{\text{PPI Home Origin Count}_c}{\sum \text{PPI Total}} \right) \times 12,680$$

This yields statistically valid estimates for every Virginia county.¹¹ For instance, Richmond City receives roughly 756 returnees annually.

Analyzing Second Chance Job Markets

We filtered 2024 Virginia Employment Commission (VEC) and Virginia Works data to isolate sectors with low entry barriers.¹³ We ignored industries like Finance or Education that frequently exclude those with records. Instead, we focused on Second Chance Friendly NAICS codes:

1. **Construction (23):** High demand and skills-based labor.¹⁴
2. **Manufacturing (31-33):** Provides stable wages and shift work.¹⁶
3. **Transportation/Logistics (48-49):** High-volume hubs like the Port of Virginia.⁵
4. **Waste Management (562):** A traditional entry point for reentry.

We developed a custom Python pipeline (`analyze_jobs.py`) to process raw VEC data. This script handles non-standard UTF-16/Tab-Delimited formats to aggregate employer data by FIPS code, aligning labor stats with correctional metrics.¹³

The Opportunity Gap Index

The Gap Score quantifies the supply and demand mismatch.

Formula:

$$\text{Gap Score} = \frac{\text{Total Second Chance Jobs}}{\text{Est. Annual Returnees}}$$

The platform classifies scores into three distinct categories:

1. **Score < 1.0 (Reentry Desert):** A structural lack of jobs requiring critical policy intervention.¹⁹
2. **Score 1.0 – 2.0 (Competitive Market):** A stable labor balance.
3. **Score > 2.0 (Mismatch):** Jobs exist, but transportation or credential barriers prevent access.²¹

Visualization Architecture

The Hub uses HTML5, Tailwind CSS, and D3.js to render interactive geospatial data.⁹

1. **Dual-Layer Logic:** Users can toggle between granular county heat maps and the nine GO Virginia economic zones.²⁵
2. **Dynamic Client-Side Aggregation:** A JavaScript engine calculates Region totals in real-time on page load. The script iterates through 133 localities to sum population and job counts.¹⁵ This ensures regional policy analysis remains mathematically consistent when users update county-level data.

Regional Economic Analysis and Labor Market Demand

To provide actionable insights, the Hub analyzes the Virginia economy across its diverse regions, identifying high-growth sectors suitable for the justice-involved population.

Statewide and Regional Labor Statistics

Total statewide employment for all occupations in 2024 was estimated at 4,064,640, a 2.1% increase from the previous year.²⁷ The Hub compares mean and median annual wages across the fourteen LWDAs to identify where reentrants have the greatest potential for wage growth.

Local Workforce Development Area (LWDA)	Employment (2024)	Mean Annual Wage	Median Annual Wage
Alexandria/Arlington (LWDA XII)	267,100	\$102,380	\$81,620
Northern Virginia (LWDA XI)	1,032,120	\$95,130	\$78,000
Capital Region (LWDA IX)	588,240	\$64,280	\$51,120
Hampton Roads (LWDA XIV)	748,660	\$58,540	\$46,410
Greater Roanoke (LWDA III)	159,160	\$57,010	\$46,010
New River/Mt. Rogers (LWDA II)	145,370	\$54,730	\$44,920
Shenandoah Valley (LWDA IV)	159,160	\$52,190	\$42,280
Southwest Virginia	52,960	\$53,670	\$44,230

(LWDA I)			
West Piedmont (LWDA X)	48,000	\$49,500	\$41,000

Data sources:.⁸

The data indicates a significant wage disparity between the Northern regions and the rest of the Commonwealth. For reentrants, the Hub highlights sectors with high demand but lower educational barriers. For example, Transportation and Material Moving is the top employing sector in the Shenandoah Valley (LWDA IV), while Production is dominant in the New River/Mt. Rogers region (LWDA II).⁸

Sector-Specific Projections and Second-Chance Alignment

The Hub identifies NAICS 23 (Construction), NAICS 31-33 (Manufacturing), and NAICS 48 (Transportation/Logistics) as the primary targets for reentry employment due to their high volume of openings and willingness to hire individuals with prior convictions.¹²

Occupation	Projected Annual Growth (to 2030)	Median Annual Wage (2024)	Suitability Rank
Heavy/Tractor-Trailer Truck Drivers	2,243	\$39,000	High
Electricians	2,053	\$47,800	High
Construction Laborers	892,690 (National)	\$47,070 (National)	High
HVAC Mechanics and Installers	1,399	\$47,900	High
Maintenance and	2,845	\$37,700	Medium

Repair Workers			
Production Associates	15,570 (LWDA II)	\$18.00/hr (Starting)	High

Data sources:.¹²

Research into second-chance employers reveals that companies like Eaton and Frito-Lay offer competitive wages (e.g., \$23.80/hr to \$29.41/hr) and active Second-Chance hiring policies in Virginia.¹³ The Hub integrates these employers into its Action Hub to provide a direct pipeline from the VADOC CTE programs to the private sector.

The G3 Program and Educational Incentives

The Get Skilled, Get a Job, Get Ahead (G3) program is a central pillar of the Hub's policy logic. Established on July 1, 2021, G3 is a tuition-assistance program for Virginia residents whose family income falls below 400% of the Federal Poverty Level.¹⁵

Eligibility and Financial Mechanics

G3 is a 'last-dollar' scholarship, meaning it covers the gap between federal/state aid and the total cost of tuition, fees, and books.³² For justice-involved individuals, who often qualify for the maximum Pell Grant, the program provides an additional Student Support Incentive Grant (SSIG) of up to \$900 per term for full-time attendance.³³

Household Size	G3 Income Threshold (400% of Poverty)
1	\$58,320
2	\$78,880
3	\$99,440
4	\$120,000

5	\$140,560
---	-----------

Data sources:.³³

Targeted Career Pathways

The G3 program focuses on five high-demand fields: Healthcare, Information Technology, Manufacturing and Skilled Trades, Early Childhood Education, and Public Safety.¹⁵ The Hub maps these programs across all 23 community colleges, identifying specific certifications like Fiber Optic Installer at Laurel Ridge or Chemical Process Operator at Mountain Empire.³³

A critical component of the G3 logic is its stackability. Students can earn a short-term credential (6-24 weeks), enter the workforce, and return within one year to continue their Associate Degree while remaining eligible for G3 funds.³² This minimizes the opportunity cost for reentrants who need immediate income but seek long-term career advancement.

The Network2Work Peer-Navigator Framework

Technical training alone is often insufficient for individuals facing the feedback loop of mass incarceration and neighborhood disadvantage.³⁶ The Hub incorporates the Network2Work (N2W) model, which received the 2021 State Innovation of the Year award for its three-network approach.¹⁶

Connectors and Navigators

The N2W methodology avoids traditional recruitment in favor of a "Ground Game" using "Connectors"—trusted neighbors and community leaders who "know everybody".¹⁶ These individuals identify job-seekers and use the N2W technology platform to match them with "Job Tickets" that describe roles in accessible language.¹⁷

"Navigators" serve as sources of ongoing encouragement, checking in with job-seekers weekly via text or phone.³⁷ This human-centric model is designed to catch problems before they lead to job loss or supervision violations, acting as an "Early Warning System" for the network.³⁷

Employment Vouching and Risk Mitigation

A primary barrier for reentrants is the "Conviction Stigma." N2W mitigates this by "Vouching" for job-seekers once they have acquired the necessary skills and addressed barriers like childcare or transportation.¹⁷ This model has demonstrated significant efficacy, with participants experiencing an average annual wage gain of 109%.¹⁷

The Danville-Green Rock CTE Partnership: A Case Study

The partnership between Danville Community College (DCC) and Green Rock Correctional Center (GROC) illustrates the Hub's philosophy of "Inside-to-Outside" alignment.³⁸ Since 2016, DCC has awarded 117 Career Studies Certificates to GROC inmates in fields like Drafting-Surveying and Air Conditioning & Refrigeration.³⁸

Programmatic Integration

VADOC facilitates this by providing classroom space and security, while DCC provides instructors and advising services.³⁸ The use of rolling enrollment schedules accommodates the unique movement patterns of the justice-impacted population, ensuring that individuals can complete their credentials regardless of when they entered the system.³⁸

The GO TEC Extension

The Hub also tracks the expansion of the GO TEC Career Connections curriculum, which introduces middle school students to manufacturing and robotics.³⁹ By mapping the GO TEC "hubs" (advanced training centers) and "spokes" (middle school labs), the Hub identifies where reentrants can access advanced-level training to complement the base skills learned at GROC.³⁹

Fiscal Logic and the Cost of Recidivism

The policy suggestions in this report are supported by a rigorous fiscal analysis of incarceration costs versus the benefits of successful reentry.

Cost of Incarceration

Virginia's Department of Corrections operates on a \$1.5 billion budget, representing a significant portion of the Commonwealth's \$31.8 billion general fund.⁴¹ The average annual cost to incarcerate a single individual is \$52,990.⁴¹ For the aging prison population (14% are 55 or older), these costs are estimated to be nearly triple due to specialized medical care.¹⁸

Cost of Recidivism

The Council of State Governments estimated the cost of recidivism in Virginia at nearly \$497 million in 2021.³ This equates to approximately \$57 per Virginia resident, ranking the Commonwealth 4th highest in the nation for per-capita recidivism costs.³

State	Cost of Recidivism per Resident (2021)
Wisconsin	\$72
Vermont	\$61
Arkansas	\$58
Virginia	\$57
Louisiana	\$50
California	\$49

Data source:.³

Research from the Justice Policy Institute indicates that aligning Virginia’s parole rates and sentence credits with comparable states could save the Commonwealth \$465 million annually.⁴¹ The Hub uses these figures to argue that every 1% reduction in the recidivism rate translates to millions of dollars in general fund savings that can be reinvested in mental health and workforce programs.¹⁸

Policy Suggestions for Economic Opportunity and Action

Based on the construction of the Hub and the underlying research, the following policy recommendations are proposed to enhance the reentry ecosystem in Virginia.

Expansion of Earned Sentence Credits and Parole Reform

The abolition of parole in 1995 has led to an aging, high-cost prison population with little incentive for personal growth.¹⁸ Expanding earned sentence credits for those who complete CTE or G3-aligned programs would not only improve prison safety by incentivizing positive behavior but also accelerate the entry of skilled labor into the Virginia market.¹⁸ Data indicates that those over 50, even with long sentences, have among the lowest recidivism rates and pose minimal risk to public safety.¹⁸

Institutionalizing the ‘No Wrong Door’ Workforce Model

The recent establishment of the Department of Workforce Development and Advancement (Virginia Works) as a central coordinating agency is a positive step.¹⁰ However, state support remains minimal at \$4.9 million.¹⁰ Policy should focus on:

1. **Scaling Opportunity Advocates:** Funding the team of Opportunity Advocates to ensure that every VADOC facility has a dedicated liaison to the Registered Apprenticeship program, which has already seen a 42% increase in active apprentices.¹⁰
2. **Universalizing G3 Access:** Removing the high school diploma/GED requirement for initial short-term credentials, allowing individuals to gain a win in the labor market while simultaneously working toward their secondary credentials through integrated education and training (IET) models.⁴²

Mitigating Spatial Mismatch through Targeted Transit and Housing

The Hub’s methodology reveals that transportation is the "Grit in the Gears" of reentry.²⁶ Policy suggestions include:

1. **"Wheels to Work" Grants:** Utilizing GO Virginia grants to fund regional transportation cooperatives that provide shuttle services from high-reentrant neighborhoods to industrial parks.¹⁴
2. **Automatic Record Sealing:** Implementing effective outreach for Virginia’s new automatic sealing laws to ensure that individuals can access housing in the "Job-Rich" Northern and Capital regions, where private landlords frequently use criminal records as an exclusionary proxy.⁴¹

Optimizing Employer Incentives: Bonding and Tax Credits

To encourage second-chance hiring, the Hub proposes a "Frictionless Incentive" model:

1. **Streamlining WOTC:** Reducing the 28-day window for WOTC applications and providing state-level tax credit matching to increase the current \$2,400 federal incentive.⁴⁴
2. **Universal Bonding Vouchers:** Automatically issuing a "Bonding Eligibility Letter" to every individual upon release from a VADOC facility, rather than requiring them to request one from a career center.⁴⁶

Leveraging Advanced Manufacturing as a Desistance Tool

Given the high success of CTE programs (9.5% recidivism), the state should incentivize private-sector manufacturers to establish training ‘Mini-Hubs’ inside facilities.¹⁹ By providing employers with a ‘Talent Preview’ of justice impacted students, the Commonwealth can transform its correctional facilities from centers of cost into engines of regional economic development.³⁸

Conclusion

The Virginia Post-Incarceration Economic Opportunity Map & Action Hub represents a paradigm shift in how the Commonwealth approaches the intersection of public safety and economic growth. By moving beyond descriptive statistics to predictive geospatial modeling, the Hub provides a blueprint for a more resilient, inclusive labor market. The 17.6% recidivism rate is not a terminal achievement but a baseline for further innovation.¹ Through the integration of the G3 educational pathway, the Network2Work human capital framework, and rigorous spatial analysis, Virginia can bridge the divide between incarceration and opportunity. The fiscal and moral imperative is clear: by investing in the desistance of its justice-involved residents, the Commonwealth secures the economic future of all its citizens.

Works cited

1. **Virginia Department of Corrections (VADOC).** (2025). VADOC Recidivism: FY2020 State Responsible Releases.
2. **Virginia Department of Corrections (VADOC).** (2025). Virginia Leads United States with Lowest Recidivism Rate.
3. **Council of State Governments Justice Center.** (2022). The Cost of Recidivism: High Price States Pay for Supervision Violations.
4. **Virginia Department of Housing and Community Development (DHCD).** (n.d.). GO Virginia Regions and State Board Resources.
5. **Virginia Economic Development Partnership (VEDP).** (2019). GO Virginia Regions and Marketing Region GIS Datasets.
6. **Virginia Works.** (2024). Community Profiles and Local Workforce Development Area (LWDA) Statistics.
7. **Center for Regional Analysis, George Mason University.** (n.d.). GO Virginia Initiative: Baseline Data and Regional Profiles.
8. **Virginia Works.** (2024). Virginia Occupational Employment and Wage Statistics: Regional Rankings.
9. **Governor Glenn Youngkin's Press Office.** (2025). Governor Glenn Youngkin Announces Virginia's Job Market Remains Resilient.
10. **Virginia Works.** (2025). Report on the Department of Workforce Development and Advancement (RD30).
11. **13News Now.** (2025). Virginia leads nation with lowest recidivism rate, report says.
12. **Bureau of Labor Statistics.** (2024). Construction: NAICS 23 Workforce Statistics and Wage Trends.
13. **ZipRecruiter.** (n.d.). Second Chance For Felons Jobs in Virginia (NOW HIRING).
14. **Bureau of Labor Statistics.** (2025). Business Employment Dynamics in Virginia: Gross Job Gains and Losses.
15. **Virginia's Community Colleges.** (n.d.). Enroll in a G3 Program in Three Simple Steps.
16. **Virginia Career Works.** (n.d.). Network2Work Overview: A Framework for Economic Mobility.
17. **Network2Work USA.** (n.d.). Peer-Navigator Framework, Technology Platform, and Poverty Reduction Results.
18. **Justice Policy Institute.** (2022). Smart Justice: Why Virginia Must Defend Earned Sentence Credits.
19. **Virginia Department of Corrections (VADOC) News.** (2025). Recidivism Data Highlights Power of Education in Correctional Facilities.
20. **Journal of Artificial Societies and Social Simulation (JASSS).** (2014). Dynamic Stock-Flow Modeling in Complex Socio-Economic Systems.

21. **World Bank Documents.** (n.d.). Migration Projections: Baseline Profiles and Statistical Registration.
22. **KNOMAD.** (2022). Stocktaking of Migration Data and Catalog for Projections.
23. **World Bank Documents.** (n.d.). Migration Projections: The Economic Case and GDP Analysis.
24. **National Center for Biotechnology Information (NCBI).** (2017). Spatial Mismatch and Reentry From Prison: PubMed Central.
25. **Roddy, J., et al.** (2024). Spatial Mismatch and Unemployment for People with a Criminal History.
26. **Sugie, N. F., & Lens, M. C.** (2017). Daytime Locations in Spatial Mismatch, Job Accessibility, and Employment at Reentry.
27. **Virginia Works.** (2024). Virginia Occupational Employment and Wage Statistics: Statewide Averages.
28. **Virginia Employment Commission (VEC).** (2024). Virginia Job Outlook Projected to 2024: Occupational Trends.
29. **Bureau of Labor Statistics.** (2025). Employment Projections News Release: 2024 to 2034.
30. **Indeed.** (n.d.). Second Chance Jobs: Employment and Hiring in Virginia.
31. **Indeed.** (n.d.). Second Chance Employment: Machine Operator and Retail Careers.
32. **Blue Ridge Community College.** (n.d.). G3 FAQ: Residency and Eligibility Rules.
33. **Laurel Ridge Community College.** (n.d.). G3 Programs: Advanced Manufacturing and Skilled Trades.
34. **Central Virginia Community College.** (n.d.). G3 – Get a Skill, Get a Job, Get Ahead.
35. **Mountain Empire Community College.** (n.d.). Program Finder and Workforce Solutions.
36. **National Center for Biotechnology Information (NCBI).** (2014). Incarceration, Prisoner Reentry, and Communities: A Feedback Loop.
37. **Piedmont Virginia Community College.** (n.d.). Network2Work@PVCC: Become a Volunteer Navigator.
38. **Virginia Department of Corrections (VADOC).** (2025). VADOC Partners with Danville Community College to Enhance CTE for Incarcerated Individuals.
39. **Institute for Advanced Learning and Research (IALR).** (n.d.). GO TEC: Career Connections Curriculum and Talent Pathways.
40. **Institute for Advanced Learning and Research (IALR).** (n.d.). Strategic Talent Pathways Starting in Middle School.
41. **R Street Institute.** (2022). Virginia's Criminal Justice Crossroads: Post-Conviction Issues and Over-Incarceration Costs.
42. **Virginia Works.** (2024). Adult Education Program Catalog and Workforce Ecosystem.
43. **Virginia Beach Sun.** (1998). Full text of "Wheels to Work" Regional Initiative.
44. **Virginia Works.** (n.d.). Work Opportunity Tax Credit (WOTC): Eligibility and Application.

45. **The Manufacturing Institute.** (n.d.). Second Chance Hiring Resource Guide: WOTC and Federal Bonding.
46. **Virginia Department of Corrections (VADOC).** (2025). Virginia Bonding Program: Overview and Free Fidelity Bonds.
47. **Virginia Department of Corrections (VADOC).** (n.d.). Virginia Federal Bonding Program (Video Guide).
48. **Correctional News.** (2025). Virginia Reports Nation's Lowest Recidivism Rate at 17.6%.